

Nephrotic Syndrome

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What is Nephrotic Syndrome?

- ☐ A kidney disorder causing damage to glomeruli → massive protein loss in urine ($>3.5\text{g/day}$)
- ☐ Leads to hypoalbuminemia, edema, hyperlipidemia, and increased infection risk
- ☐ Common in children and adults; minimal change disease is the most frequent cause

Pathophysiology: How It Happens

- □ Glomerular filtration barrier becomes abnormally permeable
- □ Albumin and other proteins leak into urine → proteinuria
- □ Loss of albumin lowers plasma oncotic pressure → fluid shifts into tissues → edema
- □ Liver compensates by increasing lipid production → hyperlipidemia

Key Clinical Signs & Symptoms

- ☐ Massive proteinuria → frothy, dark urine
- ☐ Periorbital and dependent edema (notably in mornings)
- ☐ Weight gain from fluid retention
- ☐ Hypoalbuminemia confirmed by blood tests
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- ☐ Possible hypertension and decreased urine output

Causes of Nephrotic Syndrome



□ Primary kidney diseases:

- oMinimal Change Disease (most common in children)
- oFocal Segmental Glomerulosclerosis
- oMembranous Nephropathy

□ Secondary causes:

- Diabetes Mellitus
- Lupus Erythematosus
- Infections (Hepatitis B/C)
- Amyloidosis

Diagnostic Workup

- Urinalysis: 3+ or greater proteinuria on dipstick; 24-hour urine protein quantification
- Blood tests: serum albumin (low), cholesterol (high), kidney function tests
- Kidney biopsy if diagnosis unclear or to guide treatment

Treatment Overview

- First-line: corticosteroids (e.g., prednisone) to reduce inflammation
- Diuretics to manage edema
- Antihypertensives for blood pressure control
- Lipid-lowering agents for hyperlipidemia
- Monitor and prevent infections due to immunosuppression

Nursing Care Priorities

- Monitor daily weights and strict intake/output to assess fluid balance
- Assess for signs of infection (fever, lethargy, abdominal pain)
- Observe for respiratory distress from fluid overload or ascites
- Educate patient/family on medication adherence and infection prevention
- Support nutrition and monitor for fatigue and appetite changes

Complications to Watch For

- Increased risk of serious infections (peritonitis, respiratory infections)
- Thromboembolism due to loss of anticoagulant proteins
- Relapse of nephrotic syndrome requiring long-term steroid therapy
- Potential progression to chronic kidney disease

Summary & Takeaway

- Nephrotic syndrome is a complex kidney disorder with hallmark proteinuria and edema
- Early recognition and treatment improve outcomes
- Nurses play a critical role in monitoring, educating, and supporting patients
- Stay vigilant for complications and advocate for comprehensive care

Thank You

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